

THILAK RAJ MURUGAN, MS

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Summary

Backend Software Engineer with **5+ years** of experience building **distributed systems** and **AI infrastructure** at scale.

Technical Skills

Languages: Go, Python, C++, JavaScript, TypeScript, PHP, SQL
Backend: GCP, Docker, Kubernetes, MySQL, Redis, Terraform, CI/CD, APIs, gRPC, DataDog, Microservices
Data Infra: HTAP (TiDB), Cassandra, Kafka, PySpark, HDFS, BigQuery, NoSQL
AI/ML: PyTorch, LLMs, Numpy, vLLM; **Frontend:** React.js, Redux
Coursework: Big Data, Distributed Systems, ML Infrastructure, Advanced Databases, OS

Experience

Vimeo

June 2023 – August 2024

Software Engineer 3 - IAM Platform, User Growth Team

Bengaluru, India

- Owned a high-throughput **user recommendation service** serving **10M+** users by designing backend APIs using PHP, MySQL, Memcached, Docker, **GCP** and Datadog, thus increasing free-to-paid user conversion by **20%**.
- Optimized **RBAC** based **authorization systems** by improving MySQL query performance and data access patterns.
- Modernized legacy web pages by rebuilding frontend components with React.js and TypeScript, improving logged-in user experience and boosting user interaction time by **30%**.
- Ideated and built a **video recommendation service** using **Hugging Face's** Sentence transformer based **deep learning** model, **Python** and **Golang** for the Vimeo Jam hackathon.

PropertyGuru Group

July 2021 – May 2023

Software Engineer - Fullstack

Bengaluru, India

- Built and scaled multiple **backend microservices** using **Nest.js**, **Node.js**, and **Symfony (PHP)**, serving **11M+** users across the property-tech industry in South-East Asia.
- Reduced config API latency by **25%** with Redis cache and automated customer cache invalidation, thereby increasing the efficiency of non-engineering teams.
- Built frontend webpages/micro-frontends with TypeScript, React.js, Redux and Material UI. Improved code maintainability by writing **unit tests** via Jest.

Hiver

May 2019 – June 2021

Software Engineer

Bengaluru, India

- Led the **payment platform** overhaul, driving **3× ARR growth** for the Gmail-based shared inbox product.
- Automated & optimized payment processing using the **Stripe SDK** across **Python** and **PHP** services, leveraging Fast API, webhooks, **GCP** and **RabbitMQ**, and implementing multi-factor authentication for EU transactions.

American Family Insurance

May 2025 – August 2025

MLOps Intern - AI Infrastructure Team

Madison, United States

- Engineered an **AI Inference platform** on **GCP** that autoscales inference across **1M+** BigQuery rows, eliminating infrastructure provisioning friction for AI engineers.
- Developed Infrastructure as Code (IaC) with **Terraform**, GitLab CI/CD, **Docker** and **Python** to provision GCP resources, thus automating the end-to-end deployment pipeline and reducing deployment time by **80%**.

Projects

Distributed KV Store: [Code](#)

- Built** a distributed, fault-tolerant **key-value store in Go** with horizontal scalability via sharding, **Raft** based **replication** for strong consistency, and durable storage using bbolt; achieved throughput of **23K ops/sec**.

Scalable Data Infrastructure for Analytics:

- Designed and implemented a scalable **data pipeline** for batch (BigQuery, Dataproc, Python) and stream (**Kafka**, **Spark Streaming**) processing on the NYC Taxi dataset, scanned **80GiB in <10s** using BigQuery's Dremel Engine.

Scaling Distributed Databases:

- Benchmarked **TiDB** using **Go**, scaled data size to **190 GiB** and analyzed system metrics (90 ms OLTP latency). Load-tested **Cassandra** via NoSQLBench, driving **200M** read/write operations with sub-millisecond latency.

Education

University of Wisconsin - Madison

September 2024 – May 2026

Master's in Data Engineering (Computer Sciences Dept.) - GPA: 3.9

Wisconsin, United States

Indian Institute of Technology Kharagpur

July 2014 – April 2019

Bachelor of Technology (Honors) in Electronics and Electrical Communication Engineering

Kharagpur, India